

ORIGINAL ARTICLE

Glycemia, Insulin Sensitivity Markers and Arterial Stiffness: The Framingham Heart Study

Steve R. Noumegni¹, Stephen P. Juraschek², Jordana B. Cohen³, Justin B. Echouffo-Tcheugui¹

BACKGROUND: There is limited data on the associations of (glycated hemoglobin A1C [HbA_{1c}], fasting plasma glucose, and 2-hour postload glucose) and insulin resistance (fasting plasma insulin, insulin-to-glucose ratio, homeostasis model assessment of insulin resistance, and quantitative insulin sensitivity check index) markers with arterial stiffness.

METHODS: We included 6617 Framingham Heart Study participants (53.8% women; 6.9% diabetes, mean age: 48±14 years). Arterial stiffness measures included natural log (Ln)-transformed carotid-femoral pulse wave velocity, augmentation index (Alx), central pulse pressure, carotid-brachial pulse wave velocity, carotid-radial pulse wave velocity, and brachial pulse pressure. Estimates of associations were adjusted for several clinical characteristics.

RESULTS: Adjusted means of log (Ln)-transformed carotid-femoral pulse wave velocity, Alx, and carotid-brachial pulse wave velocity were increasingly abnormal with higher HbA_{1c} (all $P<0.05$). Each 1% higher HbA_{1c} was associated with elevated CFPWV (odds ratio [OR], 1.24 [95% CI, 1.11–1.39]) and lower Alx (OR, 0.86 [95% CI, 0.78–0.95]). Each 10-mg/dL higher fasting plasma glucose was associated with elevated CFPWV (OR, 1.09 [95% CI, 1.05–1.13]), but lower Alx (OR, 0.91 [95% CI, 0.87–0.94]). Each 10-mg/dL higher 2-hour postload glucose was associated with elevated CFPWV (OR, 1.04 [95% CI, 1.01–1.07]), central pulse pressure (OR, 1.04 [95% CI, 1.02–1.06]), brachial pulse pressure (OR, 1.04 [95% CI, 1.02–1.06]), but lower Alx (OR, 0.94 [95% CI, 0.91–0.96]). Higher fasting plasma insulin, insulin-to-glucose ratio, homeostasis model assessment of insulin resistance, and lower quantitative insulin sensitivity check index were associated with higher CFPWV and lower Alx (all $P<0.05$).

CONCLUSIONS: In community-dwelling individuals, worse glycemia and insulin sensitivity were associated with abnormal arterial stiffness. (*Hypertension*. 2026;83:00–00. DOI: 10.1161/HYPERTENSIONAHA.125.26375.) • **Supplement Material.**

Key Words: cardiovascular diseases ■ fasting ■ homeostasis ■ insulin ■ risk factor

Type 2 diabetes is a major risk factor for cardiovascular disease (CVD).¹ However, the pathways linking type 2 diabetes to CVD are incompletely understood.^{2,3} Arterial stiffness has emerged as a known contributor to CVD.⁴ Emerging data suggest that type 2 diabetes is a potentially key driver of arterial stiffness,⁵ a precursor of hypertension,⁶ with the latter often coexisting with diabetes. However, there is a paucity of data on the link between type 2 diabetes and arterial stiffness.⁵ Existing population-based studies have not comprehensively assessed how various glycemic markers relate to measures of arterial stiffness, among individuals with and

without diabetes.⁵ Moreover, these studies have seldom or not explored the relation between insulin sensitivity markers and measures of arterial stiffness.⁵

Using data from the community-based FHS (Framingham Heart Study) including individuals with and without diabetes, we evaluated the associations of glycemic (fasting plasma glucose [FPG], 2-hour postload glucose [2hPG], and glycated hemoglobin [HbA_{1c}]) and insulin sensitivity (fasting plasma insulin [FPI], insulin to glucose ratio [IGR], Homeostasis Model Assessment 2 (HOMA2) of insulin resistance [HOMA-IR], and quantitative insulin sensitivity check index [QUICKI]) markers with measures

Correspondence to: Justin B. Echouffo-Tcheugui, Johns Hopkins University School of Medicine, 5501 Hopkins Bayview Cir, Baltimore, MD 21224. Email jechouf1@jhmi.edu

Supplemental Material is available at <https://www.ahajournals.org/doi/suppl/10.1161/HYPERTENSIONAHA.125.26375>.

© 2026 American Heart Association, Inc.

Hypertension is available at www.ahajournals.org/journal/hyp

NOVELTY AND RELEVANCE

What Is New?

There is a paucity of population-based data on the role of abnormal glycemic regulation in arterial stiffening. We investigated the associations of glycemic and insulin sensitivity markers with indices of arterial stiffness, including measures of central and peripheral arterial stiffness.

What Is Relevant?

Arterial stiffness measures were abnormal among individuals with higher glycemic and insulin sensitivity markers, with particularly strong associations for carotid-femoral pulse wave velocity and augmentation index.

Clinical/Pathophysiological Implications?

Our findings suggest that aggressively addressing diabetes and the related insulin resistance may lead to better vascular health, and consequently fewer cardiovascular disease events.

Nonstandard Abbreviations and Acronyms

2hPG	2-hour postload blood glucose
AIx	augmentation index
BPP	brachial pulse pressure
CBPWV	carotid-brachial pulse wave velocity
CFPWV	carotid-femoral pulse wave velocity
CPP	central pulse pressure
CRPWV	carotid-radial pulse wave velocity
CVD	cardiovascular disease
FHS	Framingham Heart Study
FOS	Framingham Offspring Study
FPG	fasting plasma glucose
FPI	fasting plasma insulin
GLP-1RA	glucagon-like peptide-1 receptor agonist
HbA_{1c}	glycated hemoglobin
HOMA2	Homeostatic Model Assessment 2
IGR	Insulin-to-glucose ratio
OR	odds ratio
QUICKI	quantitative insulin sensitivity check index
SGLT2	sodium-glucose transport protein 2

of arterial stiffness. The latter include measures of central vasculature arterial stiffness (carotid-femoral pulse wave velocity [CFPWV], augmentation index [AIx], and central pulse pressure [CPP]), and peripheral vasculature arterial stiffness (carotid-brachial pulse wave velocity [CBPWV], carotid-radial pulse wave velocity [CRPWV], and brachial pulse pressure [BPP]).

METHODS

Study Sample

The FHS study design and methods have been published previously.^{7,8} This cross-sectional investigation focused on FHS

participants who attended their seventh exam (1998–2001) for the FOS (Framingham Offspring Study cohort, generation 2, exam 7) or the first exam (2002–2005) for the third generation (Gen 3 cohort, generation 3, exam 1),^{7,8} as attendees at these examinations underwent arterial tonometry.

Of the 7487 eligible participants, we excluded 870 without any arterial tonometry data or without any concomitant glycemic markers (n=849 FOS cohort and n=21 Gen 3 cohort), giving us a main final sample of 6617 participants. Only participants without diabetes at the index examinations (n=6162) were included in the analyses exploring the association between insulin sensitivity markers and arterial stiffness measures.

All data from the FHS cohorts underlying this article are publicly available at the National Heart, Lung, and Blood Institute Biologic Specimen and Data Repository Information Coordinating Center—at <https://biolincc.nhlbi.nih.gov/studies/fhs/>.

The institutional review board at Boston University approved the study protocol. All FHS participants provided written informed consent.

Assessment of Glycemic and Insulin Sensitivity Markers

FPG level was measured with a hexokinase reagent kit (A-gent glucose test, Abbott, South Pasadena, CA). HbA_{1c} was measured by high-performance liquid chromatography. The laboratory's nondiabetic mean±SD for this assay was 5.00±0.65%, and the inter- and intraassay coefficients of variation were <3%. The 2hPG was measured 2 hours after administration of a standard 75 g oral glucose load in subjects not known to have diabetes. Diabetes was defined as an FPG ≥126 mg/dL, or nonfasting glucose ≥200 mg/dL, or HbA_{1c} ≥6.5%, or use of hypoglycemic medications at the index clinical examination.⁹ Prediabetes was defined as an FPG of 100 to 125 mg/dL or HbA_{1c} of 5.7% to 6.4%.⁹

FPI levels were measured in plasma as total immunoreactive insulin and were standardized to serum levels for reporting purposes. IGR was obtained as the ratio of FPI to FPG. HOMA-IR was calculated using the formula: (FPG [mmol/L]×FPI [μU/mL])/22.5.¹⁰ QUICKI was estimated using the formula: 1/(log [FPI, μU/mL] + log [FPG, mg/dL]).¹¹ A higher FPI,

IGR, and HOMA-IR, and a lower QUICKI index are all indicative of lower insulin sensitivity or greater insulin resistance. These indices are excellent surrogates for predicting insulin sensitivity as measured by the gold standard hyperinsulinemic euglycemic glucose clamp.¹² We calculated HOMA of Beta Cell Function using the formula: $(20 \times \text{FPI} [\mu\text{IU/mL}]) / (\text{FPG} [\text{mmol/L}] - 3.5)$.¹⁰ In addition, HOMA2 for insulin resistance, HOMA2 for β -cell function, and HOMA2 for insulin sensitivity were estimated with the HOMA2 Calculator, version 2.2.3 (Diabetes Trials Unit, University of Oxford, Oxford, United Kingdom).^{10,13}

Assessment of Arterial Stiffness

Applanation tonometry on the right brachial, radial, femoral, and carotid arteries was performed by trained sonographers on participants in supine position using a standardized protocol with excellent reproducibility (intraclass correlation coefficients of 0.93–0.95 for repeated measures and inter-observer correlations of ≥ 0.97).^{14,15} Arterial waveforms were signal-averaged using the electrocardiographic R wave as the fiducial point. The brachial arterial waveform was integrated and calibrated with the systolic and diastolic auscultatory blood pressure obtained during tonometry to derive mean arterial pressure. Calibrated carotid pressure was used as a surrogate for central pressure.¹⁴ CPP was defined as the difference between the peak and trough of the calibrated carotid pressure waveform. BPP was defined as the difference between the peak and trough of the calibrated brachial pressure waveform. Alx was computed from the carotid pressure waveform as described previously.¹⁴ The heart rate-corrected Alx was calculated using the formula: $\text{Alx} - 0.39 \times (75 - \text{heart rate})$.¹⁶

CFPWV, CBPWV, and CRPWV values were calculated from tonometry waveforms and body surface measurements, which were adjusted for parallel transmission in the brachiocephalic artery and aortic arch with the use of the suprasternal notch as a fiducial point.¹⁴ The carotid-femoral transit path spans the aorta, making CFPWV a measure of aortic stiffness (systemic or central arterial stiffness), as are Alx and CPP. In contrast, the carotid-brachial transit path spans the subclavian and brachial arteries, while the carotid-radial transit path spans the subclavian, brachial, and radial arteries, making CBPWV and CRPWV measures of muscular artery stiffness (peripheral arterial stiffness), as is BPP.¹⁴ For each measure of arterial stiffness, elevated values were defined as ≥ 75 th percentile of the distribution, and overall arterial stiffness was defined by at least one of the 6 measures ≥ 75 th percentile of the distribution. Central arterial stiffness was defined by at least 1 value of CFPWV, Alx, or CPP ≥ 75 th percentile of the distribution, while peripheral arterial stiffness was defined by at least 1 value ≥ 75 th percentile of the distribution among CBPWV, CRPWV, or BPP.

Our primary stiffness outcome or measure was CFPWV, and all other arterial stiffness measures were considered as secondary outcomes.

Covariates

Blood pressure, height, and weight measurements, as well as data on CVD history, antihypertensive medications, diabetes medications, smoking history, and laboratory tests, were obtained at the scheduled index examinations. Brachial blood pressure was measured using a standard mercury column sphygmomanometer and a cuff of appropriate size. The

average of the 2 physician-obtained measurements on the right arm of the seated participants was used as the blood pressure. Body mass index was defined as weight divided by the square of height (kg/m^2). Current smoking status was defined as smoking at least 1 cigarette a day for the year preceding index examination. Serum creatinine was measured using the modified Jaffe method. Circulating total cholesterol and high-density lipoprotein cholesterol concentrations were measured using enzymatic methods.

Statistical Analyses

The participants' characteristics are presented as frequency (percentage) for categorical variables and mean $\pm$ SD or median (25th–75th percentiles) for continuous variables where applicable. They were compared based on diabetes status using the χ^2 test, the Student *t* test, or equivalents where appropriate. Because CFPWV, HbA_{1c} , FPG, and 2hPG distributions were left-skewed, we performed a logarithmic transformation of the results; therefore, the values considered in linear regression analyses were $\text{Ln}(\text{CFPWV}, \text{m/s})$, $\text{Ln}(\text{HbA}_{1c}, \%)$, $\text{Ln}(\text{FPG}, \text{mg/dL})$, and $\text{Ln}(\text{2hPG}, \text{mg/dL})$. To keep consistency with prior studies,^{14,15,17} we also inverse-transformed CFPWV ($1/\text{CFPWV}$) to reduce heteroscedasticity and then multiplied the value by -1000 to restore directionality. The resulting variable was referred to as inverse CFPWV throughout the article.

We conducted multivariable regression analyses to model the associations of glycemic markers (HbA_{1c} , FPG, and 2hPG) and insulin sensitivity markers (FPI, IGR, HOMA-IR, and QUICKI) with measures of arterial stiffness (CFPWV, Alx, CPP, CBPWV, CRPWV, and BPP). First, we evaluated the associations between these markers as continuous variables and measures of arterial stiffness as continuous variables using linear regression models and reporting the beta coefficient (β) estimates. Second, we evaluated the associations of categorical glycemic and insulin sensitivity markers with continuous measures of arterial stiffness using linear regression models and reporting adjusted means. Third, we evaluated the associations between continuous and categorical glucose dysregulation markers with elevated CFPWV, elevated Alx, elevated CPP, elevated CRPWV, and elevated BPP using logistic regression models. The categorization of glycemic markers was done using clinically relevant cut points as recommended by the American Diabetes Association.⁹ The categorization of insulin sensitivity markers was done using quartiles' distributions.

We also explored the association of diabetes status (diabetes versus no-diabetes) and glycemic status (diabetes/prediabetes versus normal glycemia) with continuous arterial stiffness measures reporting adjusted means and with abnormal arterial stiffness measures (elevated values ≥ 75 th percentiles) and with peripheral arterial stiffness (binary variable), central arterial stiffness (binary variable), and overall arterial stiffness (binary variable), reporting odds ratios (ORs). The analyses of insulin sensitivity markers (as exposures) were conducted among participants without diabetes. For each linear regression model used, we plotted the residuals to assess model assumptions.

All regression models were adjusted for age, sex, cohort (FOS versus Gen3), body mass index, plasma creatinine, SBP, use of antihypertensive medications, smoking status, total to high-density lipoprotein cholesterol ratio, and history of CVD. Regression coefficients (β and ORs) were reported with their 95% CI and adjusted means with standard errors.

Given the well-known sexual dimorphism in vascular aging and diabetes,¹⁸ we additionally conducted a priori sex-stratified analysis on the association between categorical glycemic markers and continuous measures of arterial stiffness. We also conducted sensitivity analyses to assess the robustness of the primary findings. These analyses included repeating the linear regression models with categorical glycemic markers as exposures and using untransformed CFPWV and heart rate-corrected Alx as outcomes, with adjusted means reported. We also evaluated additional markers of insulin production (HOMA of Beta Cell Function and HOMA2 for β -cell function) and insulin sensitivity (HOMA2 for insulin resistance and HOMA2 for insulin sensitivity) as continuous and categorical exposures in relation to continuous measures of arterial stiffness.

Data were analyzed using R, version 4.3.2. A 2-sided *P* value of <0.05 was considered statistically significant.

RESULTS

Characteristics of Study Participants

A total of 6617 participants were included in our study (mean age 48.3±13.7 years, 53.8% women). Of all participants, 455 had diabetes (6.9%) at the index examination, 760 had prediabetes (11.5%), and 5385 had normal glycemia (81.6%).

The baseline characteristics of participants by diabetes status are summarized in Table 1. Participants with diabetes were older and more likely to be men, with higher body mass index and serum creatinine, lower high-density lipoprotein cholesterol, and a higher prevalence of hypertension and CVD. Except for Alx, measures of arterial stiffness were higher in participants with diabetes than in those without diabetes. Similarly, apart from Alx, the prevalence of abnormal values in the measures of arterial stiffness was higher in participants with diabetes or prediabetes than with normal glycemia.

HbA_{1c} and Arterial Stiffness

Each 1-unit increase in HbA_{1c} (Ln [HbA_{1c}]) was associated with an increase in Ln (CFPWV; β , 0.14 [95% CI, 0.10–0.18]) and a decrease in Alx (β , −5.11 [95% CI, −7.96 to −2.26]; Table S1). compared with participants with HbA_{1c} <5.7%, those with HbA_{1c} ≥6.5%, had significantly higher adjusted means of Ln (CFPWV; Table 2). Compared with participants with HbA_{1c} <5.7%, those with HbA_{1c} of 5.7% to 6.4% and HbA_{1c} ≥6.5% had lower adjusted means of Alx (Table 2).

Each 1% change in HbA_{1c} was associated with higher adjusted odds of having elevated CFPWV (OR, 1.24 [95% CI, 1.11–1.39]), elevated CBPWV (OR, 1.08 [95% CI, 1.01–1.18]), and lower odds of having elevated Alx (OR, 0.86 [95% CI, 0.78–0.95]; Table S2; Figure). Compared with participants with HbA_{1c} <5.7%, those with HbA_{1c} ≥6.5% had higher odds of having elevated CFPWV (OR, 1.90 [95% CI, 1.35–2.67]) and lower odds of having elevated Alx (OR, 0.62 [95% CI, 0.46–0.83]; Table 3). There

was no statistically significant difference in CPP, CRPWV, and BPP, and in proportions of abnormalities in these indices across HbA_{1c} categories (all *P*>0.05, Tables 2 and 3).

FPG and Arterial Stiffness

Each unit increase in FPG (modeled as Ln [FPG]) was associated with an increase in Ln (CFPWV; β , 0.13 [95% CI, 0.10–0.16]), CBPWV (β , 0.35 [95% CI, 0.08–0.63]), and in CRPWV (β , 0.25 [95% CI, 0.01–0.49]), but a decrease in Alx (β , −6.46 [95% CI, −8.53 to −4.40; Table S1). Compared with participants with FPG <100 mg/dL, those with FPG of 100 to 125 mg/dL and those with FPG ≥126 mg/dL had significantly higher adjusted means of Ln (CFPWV), CBPWV, and CRPWV, and significantly lower adjusted means of Alx (Table 2).

Each 10 mg/dL change in FPG was associated with higher adjusted odds of having elevated CFPWV (OR, 1.09 [95% CI, 1.05–1.13]), elevated CBPWV (OR, 1.04 [95% CI, 1.01–1.07]), and lower odds of having elevated Alx (OR, 0.91 [95% CI, 0.87–0.94]; Table S2; Figure). Compared with participants with FPG <100 mg/dL, those with FPG ≥126 mg/dL had higher odds of having elevated CFPWV (OR, 2.85 [95% CI, 1.93–4.23]), elevated CBPWV (OR, 1.47 [95% CI, 1.11–1.94]), and elevated BPP (OR, 1.37 [95% CI, 1.02–1.84]). Compared with participants with FPG <100 mg/dL, those with FPG of 100 to 125 mg/dL (OR, 0.81 [95% CI, 0.69–0.96]), and those with FPG ≥126 mg/dL (OR, 0.44 [95% CI, 0.31–0.61]) had lower odds of having elevated Alx (Table 3). There was no significant difference in CPP across FPG categories (all *P*>0.05, Tables 2 and 3).

2hPG and Arterial Stiffness

Increasing 2hPG was positively associated with CFPWV, CPP, CBPWV, and BPP and negatively associated with Alx (Table S1). Across the categories of worsening 2hPG, we observed progressively abnormal adjusted mean values of Ln (CFPWV) and BPP (higher values) and Alx (lower values; all *P*_{Trend} <0.001; Table 2).

Each 10 mg/dL change in 2hPG was associated with higher adjusted odds of having elevated CFPWV (OR, 1.04 [95% CI, 1.01–1.07]), elevated CPP (OR, 1.04 [95% CI, 1.02–1.06]), elevated BPP (OR, 1.04 [95% CI, 1.02–1.06]), and lower odds of having elevated Alx (OR, 0.94 [95% CI, 0.91–0.96]; Table S2; Figure). Compared with participants with 2hPG <140 mg/dL, those with 2hPG ≥200 mg/dL had higher odds of elevated BPP (OR, 1.62 [95% CI, 1.03–2.52]). Compared with participants with 2hPG <140 mg/dL, those with 2hPG of 140 to 199 mg/dL (OR, 0.65 [95% CI, 0.49–0.86]) and those with 2hPG ≥200 mg/dL (OR, 0.27 [95% CI, 0.15–0.47]) had lower odds of elevated Alx (Table 3). There were no significant differences in CBPWV and CRPWV across 2hPG categories (all *P*>0.05; Tables 2 and 3).

Table 1. Characteristics of Participants by Diabetes Status

	Overall	No diabetes	Diabetes	P value
N	6617	6162	455	
General characteristics				
Age, y	48.26 (13.66)	47.40 (13.36)	59.95 (12.11)	<0.001
Men, n (%)	3057 (46.2)	2801 (45.5)	256 (56.3)	<0.001
Cohort				<0.001
Offspring cohort, n (%)	2574 (38.9)	2242 (36.4)	332 (73.0)	
Gen 3 cohort, n (%)	4043 (61.1)	3920 (63.6)	123 (27.0)	
Glycemic status (%)				<0.001
Normal glycemia	5385 (81.6)	5385 (87.6)	0 (0.0)	
Prediabetes	760 (11.5)	760 (12.4)	0 (0.0)	
Diabetes	455 (6.9)	0 (0.0)	455 (100.0)	
Diabetes medication, n (%)	256 (3.9)	0 (0.0)	256 (56.3)	<0.001
BMI, kg/m ²	27.38 (5.53)	27.04 (5.26)	32.01 (6.83)	<0.001
Serum creatinine, mg/dL	0.83 (0.22)	0.82 (0.21)	0.88 (0.32)	<0.001
Systolic blood pressure, mm Hg	120.96 (17.13)	120.06 (16.64)	133.05 (18.97)	<0.001
Diastolic blood pressure, mm Hg	74.89 (9.80)	74.90 (9.72)	74.75 (10.80)	0.760
Hypertension, n (%)	2713 (41.0)	2429 (39.4)	284 (62.4)	<0.001
Hypertension medication, n (%)	1225 (18.5)	971 (15.8)	254 (55.8)	<0.001
Current smokers, n (%)	978 (14.8)	922 (15.0)	56 (12.3)	0.140
Total cholesterol, mg/dL	193.33 (36.45)	193.68 (36.21)	188.48 (39.37)	0.003
HDL cholesterol, mg/dL	54.35 (16.56)	54.98 (16.53)	45.70 (14.48)	<0.001
Total/HDL cholesterol ratio	3.87 (1.39)	3.83 (1.36)	4.47 (1.61)	<0.001
Lipid-lowering medication, n (%)	806 (12.2)	628 (10.2)	178 (39.2)	<0.001
History of cardiovascular diseases, n (%)	396 (6.0)	279 (4.5)	117 (25.7)	<0.001
Glycemic and insulin sensitivity markers				
FPG, mg/dL	95.00 (89.00–102.00)	94.00 (88.00–100.00)	138.50 (124.20–175.80)	<0.001
2hPG, mg/dL	103.00 (87.00–123.00)	102.00 (87.00–122.00)	196.50 (167.20–251.50)	<0.001
HbA _{1c}	5.41 (5.20–5.70)	5.40 (5.20–5.65)	6.87 (6.10–7.88)	<0.001
FPI, μ U/mL	6.90 (3.85–12.20)	6.38 (3.72–11.24)	16.50 (10.69–24.15)	<0.001
IGR, (μ U/mL)/(mg/dL)	0.07 (0.04–0.12)	0.06 (0.04–0.11)	0.11 (0.07–0.17)	<0.001
HOMA-IR	1.63 (0.88–2.97)	1.49 (0.84–2.69)	5.60 (3.47–9.33)	<0.001
QUICKI	0.16 (0.02)	0.16 (0.02)	0.13 (0.01)	<0.001
Measures of arterial stiffness				
CFPWV, m/s	7.37 (6.39–8.88)	7.25 (6.34–8.65)	10.23 (8.53–12.82)	<0.001
Alx, %	9.87 (14.08)	9.84 (13.99)	10.29 (15.29)	0.525
CPP, mm Hg	51.70 (14.05)	51.20 (13.54)	58.90 (18.54)	<0.001
CBPWV, m/s	8.51 (1.82)	8.44 (1.80)	9.49 (1.85)	<0.001
CRPWV, m/s	9.71 (1.52)	9.67 (1.51)	10.27 (1.52)	<0.001
BPP, mm Hg	53.41 (12.37)	52.93 (11.91)	60.01 (16.10)	<0.001

Values are presented as mean (SD) or median (25th–75th percentiles) for continuous variables, and frequency (percentage) for categorical variables. 2hPG indicates 2-hour postload blood glucose; Alx, augmentation index; BMI, body mass index; BPP, brachial pulse pressure; CBPWV, carotid-brachial pulse wave velocity; CFPWV, carotid-femoral pulse wave velocity; CPP, central pulse pressure; CRPWV, carotid-radial pulse wave velocity; FPG, fasting plasma glucose; FPI, fasting plasma insulin; HbA_{1c}, glycated haemoglobin; HDL, high-density lipoprotein; HOMA-IR, homeostasis model assessment of insulin resistance; IGR, insulin-to-glucose ratio; and QUICKI, quantitative insulin sensitivity check index.

Glycemic Status and Arterial Stiffness

Compared with participants without diabetes, those with diabetes had significantly higher adjusted means of Ln (CFPWV), CPP, CBPWV, and BPP, and lower adjusted

means of Alx (Table S3). Compared with individuals with normoglycemia, those with prediabetes and with diabetes had higher adjusted means of Ln (CFPWV), CPP, CBPWV, and BPP and lower adjusted means of Alx (Table S3). Compared with participants without

Table 2. Adjusted Means for Continuous Measures of Arterial Stiffness by Categories of Glycemic Markers.

	Categories of HbA _{1c} %				Categories of FPG, mg/dL				Categories of 2hPG, mg/dL			
	<5.7%	5.7%–6.4%	≥6.5%	<i>P</i> _{Trend}	<100 mg/dL	100–125 mg/dL	≥126 mg/dL	<i>P</i> _{Trend}	<140 mg/dL	140–199 mg/dL	≥200 mg/dL	<i>P</i> _{Trend}
CFPWV, m/s	8.68 (0.07)	8.72 (0.08)	9.59 (0.13)	<0.001	8.50 (0.07)	8.68 (0.08)	9.76 (0.13)	<0.001	7.90 (0.07)	8.18 (0.09)	8.63 (0.15)	<0.001
Ln (CFPWV, m/s)	2.09 (0.006)	2.09 (0.007)	2.16 (0.01)	<0.001	2.06 (0.005)	2.08 (0.006)	2.18 (0.01)	<0.001	2.04 (0.008)	2.04 (0.01)	2.06 (0.02)	<0.001
Inverse CFPWV, s/m	−130.94 (0.68)	−130.52 (0.81)	−124.27 (1.19)	<0.001	−133.13 (0.67)	−131.25 (0.76)	−123.07 (1.26)	<0.001	−140.22 (1.02)	−136.65 (1.27)	−135.38 (1.98)	<0.001
Alx, %	12.11 (0.40)	12.06 (0.48)	9.74 (0.70)	0.002	11.98 (0.40)	11.17 (0.47)	6.99 (0.75)	<0.001	7.97 (0.61)	6.86 (0.76)	1.92 (1.20)	<0.001
Alx@75, %	6.89 (0.38)	6.91 (0.46)	6.01 (0.66)	0.39	6.52 (0.38)	6.57 (0.43)	3.81 (0.71)	<0.001	2.99 (0.58)	2.92 (0.73)	−1.09 (1.14)	<0.001
CPP, mmHg	52.54 (0.39)	52.39 (0.47)	53.15 (0.68)	0.56	52.38 (0.38)	52.12 (0.43)	53.75 (0.72)	0.07	50.29 (0.57)	51.98 (0.71)	50.81 (1.12)	0.01
CBPWV, m/s	8.47 (0.05)	8.47 (0.06)	8.74 (0.09)	0.01	8.38 (0.05)	8.51 (0.06)	8.71 (0.10)	<0.001	8.00 (0.08)	8.11 (0.10)	8.13 (0.15)	0.30
CRPWV, m/s	9.60 (0.04)	9.56 (0.05)	9.70 (0.08)	0.27	9.53 (0.04)	9.58 (0.05)	9.77 (0.08)	0.02	9.27 (0.07)	9.41 (0.08)	9.29 (0.13)	0.13
BPP, mmHg	53.93 (0.34)	53.71 (0.41)	54.93 (0.60)	0.14	53.67 (0.34)	53.59 (0.38)	56.09 (0.63)	<0.001	53.02 (0.49)	54.76 (0.62)	55.32 (0.98)	<0.001

Values are adjusted means (standard errors) from multivariable linear regression models with each glycemic marker as an independent categorical variable. Adjustment variables included age, sex, cohort, BMI, plasma creatinine, SBP, use of antihypertensive medications, smoking status, total to HDL cholesterol ratio, and history of cardiovascular disease. 2hPG indicates 2-hour postload blood glucose; Alx, augmentation index; BMI, body mass index; BPP, brachial pulse pressure; CBPWV, carotid-brachial pulse wave velocity; CFPWV, carotid-femoral pulse wave velocity; CPP, central pulse pressure; CRPWV, carotid-radial pulse wave velocity; FPG, fasting plasma glucose; HbA_{1c}, glycated haemoglobin; HDL, high-density lipoprotein; and SBP, systolic blood pressure.

diabetes, those with diabetes had decreased odds of having elevated Alx (OR, 0.50 [95% CI, 0.37–0.65]), and increased odds of having elevated CFPWV (OR, 2.25 [95% CI, 1.64–3.09]), elevated CPP (OR, 1.37 [95% CI, 1.05–1.77]), elevated BPP (OR, 1.63 [95% CI, 1.27–2.09]), overall central arterial stiffness (OR, 1.52 [95% CI, 1.10–2.11]), overall peripheral arterial stiffness (OR, 1.51 [95% CI, 1.16–1.97]), and overall arterial stiffness (OR, 1.51 [95% CI, 1.05–2.24]; Figure; Table S3).

Insulin Sensitivity Markers and Arterial Stiffness

In participants without diabetes, increasing levels of FPI, IGR, and HOMA-IR were positively associated with CFPWV, CBPWV, and CRPWV, and negatively associated with Alx. Increasing QUICKI was negatively associated with CFPWV, CBPWV, and CRPWV and positively associated with Alx (Tables S4 and S5). Each unit increase in FPI, IGR, and HOMA-IR, as well as greater categories of these markers, was associated with higher odds of having elevated CFPWV and lower odds of having elevated Alx (Tables S6 and S7). Each unit increase in QUICKI and greater QUICKI categories were associated with lower odds of having elevated CFPWV and higher odds of having elevated Alx (Tables S6 and S7). There were no significant differences in CPP, CBPWV, CRPWV, and BPP across categories of insulin sensitivity markers (all *P*>0.05; Tables S5 and S7).

Sensitivity Analyses

Overall, similar results were consistently obtained with natural logarithmic transformation and with inverse transformation of CFPWV (Tables 2; Tables S1, S4, and S5). The results of the association between glycemic markers and arterial stiffness measures were also consistent with the main results using heart rate-corrected Alx and untransformed CFPWV (Table 2), as well as in sex-stratified analyses (Tables S8 and S9). The results of the association between HOMA of Beta Cell Function, HOMA2 for insulin resistance, HOMA2 for β -cell function, HOMA2 for insulin sensitivity, and arterial stiffness measures were consistent with the results obtained in our main analyses (Tables S10 and S11).

DISCUSSION

In a large community-based cohort of young-to-middle-aged individuals, we studied the association of several markers of glucose dysregulation, including glycemic and insulin sensitivity markers, as well as diabetes and prediabetes status, with tonometry-based measures of arterial stiffness. Our study observed robust and consistent associations between various glucose dysregulation markers and greater arterial stiffness. Indeed, we observed that across all glycemic and insulin sensitivity markers, there were significant associations between worse levels of these measures and abnormal measures of stiffness, notably higher CFPWV and lower Alx.

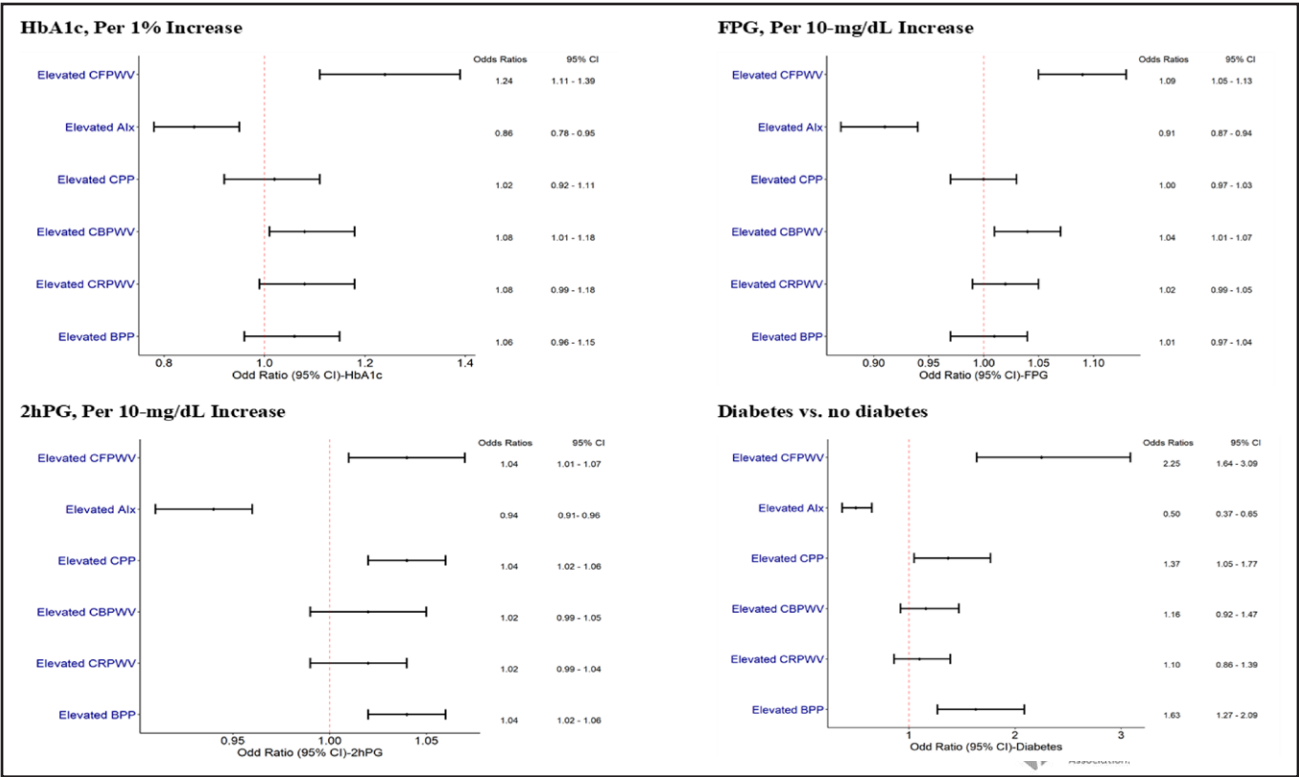


Figure. Adjusted odds ratios for categorical measures of arterial stiffness by increments in continuous glycemic markers and by diabetes status.

Adjustment variables included age, sex, cohort, body mass index, plasma creatinine, SBP, use of antihypertensive medications, smoking status, total to high-density lipoprotein cholesterol ratio, and history of cardiovascular disease. 2hPG indicates 2-hour postload blood glucose; Alx, augmentation index; BPP, brachial pulse pressure; CBPWV, carotid-brachial pulse wave velocity; CFPWV, carotid-femoral pulse wave velocity; CPP, central pulse pressure; CRPWV, carotid-radial pulse wave velocity; FPG, fasting plasma glucose; and HbA_{1c}, glycated hemoglobin.

Our observations extend the findings from prior studies. Indeed, diabetes has been previously described as a risk factor for arterial stiffness,^{19,20} although the extent to which this relation exists across the spectrum of various glycemia measures (including nondiabetic range hyperglycemia) has been less clear. In addition to assessing the association between diabetes status and arterial stiffness, studies have also reported on the association of glycemic markers and arterial stiffness, with some studies focusing on HbA_{1c},^{21,22} others on FPG,^{20,23} but few have examined all glycemic markers simultaneously, or included glucose measures obtained postload.^{20,24,25} Our findings on the positive association between HbA_{1c}, FPG and 2hPG and measures of arterial stiffness, mainly pulse wave velocity, are in keeping with those of the few prior studies that examined these markers.^{20–25} The negative associations observed between glycemic markers and Alx (or similarly for Alx @75) are also consistent with prior studies that have reported a higher Alx among individuals without diabetes compared with those with diabetes,^{26,27} which contrasts with common assumptions. It has been suggested that this may be due to mathematical limitations of the Alx formula, which may not always accurately reflect physiological changes in the central vasculature.²⁸ Moreover, a prior study reported

that individuals with an ankle-brachial index >1.5, indicative of arterial stiffness, tend to have a lower Alx than those with a lower ankle-brachial index.²⁹ Overall, prior population-based studies on glycemia and arterial stiffness did not investigate the full spectrum of tonometry measures that assess both peripheral and central arterial stiffness simultaneously.^{20–25}

Our findings of positive associations between markers of insulin sensitivity or resistance and measures of arterial stiffness are also in concordance with existing data.^{19,30–32} The few differences observed between our results and those from these prior studies could be explained by the differences in the extent of adjustment for covariates, the methods used to assess arterial stiffness, the demographic characteristics of participants, and the methods of assessment of insulin resistance. Indeed, studies such as ours that used insulin resistance measures based on dynamic tests such as the oral glucose tolerance test or hyperinsulinemic-euglycemic clamp could reveal stronger associations of insulin sensitivity with arterial stiffness than with fasting-based measures only, such as FPI, IGR, or HOMA-IR.

Our study can help further our understanding of the underlying drivers of the association between

Table 3. Adjusted OR for Categorical Measures of Arterial Stiffness by Categories of Glycemic Markers

	Categories of HbA _{1c} , %				Categories of FPG, mg/dL				Categories of 2hPG, mg/dL			
	<5.7%	5.7%–6.4%	≥6.5%	<i>P</i> _{Trend}	<100 mg/dL	100–125 mg/dL	≥126 mg/dL	<i>P</i> _{Trend}	<140 mg/dL	140–199 mg/dL	≥200 mg/dL	<i>P</i> _{Trend}
Elevated CFPWV	Ref.	0.90 (0.73–1.12)	1.90 (1.35–2.67)	<0.001	Ref.	1.12 (0.92–1.35)	2.85 (1.93–4.23)	<0.001	Ref.	1.43 (1.03–1.97)	1.16 (0.68–1.99)	0.09
Elevated Alx	Ref.	1.04 (0.88–1.22)	0.62 (0.46–0.83)	0.003	Ref.	0.81 (0.69–0.96)	0.44 (0.31–0.61)	<0.001	Ref.	0.65 (0.49–0.86)	0.27 (0.15–0.47)	<0.001
Elevated CPP	Ref.	0.89 (0.75–1.07)	1.14 (0.86–1.52)	0.20	Ref.	0.94 (0.80–1.10)	1.11 (0.81–1.51)	0.50	Ref.	1.35 (1.04–1.74)	1.30 (0.82–2.04)	0.05
Elevated CBPWV	Ref.	1.10 (0.94–1.29)	1.32 (1.02–1.71)	0.07	Ref.	1.08 (0.94–1.25)	1.47 (1.11–1.94)	0.02	Ref.	1.17 (0.91–1.49)	1.49 (0.98–2.27)	0.11
Elevated CRPWV	Ref.	1.01 (0.86–1.18)	1.26 (0.97–1.63)	0.19	Ref.	1.03 (.89–1.19)	1.28 (0.97–1.69)	0.21	Ref.	1.29 (1.01–1.64)	1.22 (0.79–1.87)	0.09
Elevated BPP	Ref.	0.97 (0.82–1.15)	1.07 (0.80–1.41)	0.81	Ref.	0.93 (0.79–1.09)	1.37 (1.02–1.84)	0.03	Ref.	1.27 (0.98–1.63)	1.62 (1.03–2.52)	0.03

Values are adjusted odds ratios from multivariable logistic regression models with each glycemic marker as an independent categorical variable. Adjustment variables included age, sex, cohort, BMI, plasma creatinine, SBP, use of antihypertensive medications, smoking status, total to HDL cholesterol ratio, and history of cardiovascular disease. 2hPG indicates 2-hour postload blood glucose; Alx, augmentation index; BMI, body mass index; BPP, brachial pulse pressure; CBPWV, carotid-brachial pulse wave velocity; CFPWV, carotid-femoral pulse wave velocity; CPP, central pulse pressure; CRPWV, carotid-radial pulse wave velocity; FPG, fasting plasma glucose; HbA_{1c}, glycated haemoglobin; HDL, high-density lipoprotein; OR, odds ratio; and SBP, systolic blood pressure.

dysglycemia and arterial stiffness. Indeed, our study attempted to capture the whole spectrum of dysglycemia (including prediabetes and diabetes), as well as different aspects of the metabolic milieu associated with diabetes that may affect arterial stiffness. These include glucotoxicity (possibly captured by FPG and HbA_{1c}), lipotoxicity/insulin resistance and hyperinsulinemia (captured by the degree of insulin resistance and fasting insulinemia), or increased tissue glycation leading to fibrosis (captured by HbA_{1c}).⁵ It is also important to mention that dysglycemia and arterial stiffness can have a bidirectional relation.^{17,33} Participants with diabetes in our cohort exhibited a higher burden of cardiometabolic risk factors, raising the possibility that a subset may also meet criteria for metabolic syndrome. However, we did not formally assess metabolic syndrome, and there was substantial heterogeneity within the diabetes group, as not all individuals had hypertension, for example. This suggests that the observed associations between glycemic/insulin resistance markers and arterial stiffness measures are not solely driven by a uniform metabolic syndrome phenotype but reflect vascular effects across a spectrum of metabolic risk.

Our results have clinical and public health implications. Regarding the clinical relevance, our findings support the notion that abnormal glucose regulation is a marker of risk of vascular stiffness, which can contribute to incident hypertension or CVD. It remains unclear whether novel medications, such as SGLT2 (sodium-glucose transport protein 2) inhibitors and GLP-1RAs (glucagon-like peptide-1 receptor agonists), especially in high-risk subgroups of patients such as individuals with diabetes, can also have a beneficial effect on arterial stiffness.³⁴ Additional potentially beneficial lifestyle interventions that include increasing physical activity levels and a healthy

diet can help address vascular stiffness in the setting of diabetes. In brief, our results might help identify adults at increased risk for vascular disease, inform screening strategies, and plan the assessment of the impact of interventions targeting glycemia on vascular stiffness.

The strengths of our study are manifold. These include a well-characterized large community-based sample of individuals, the availability of several markers of glycemic dysregulation, and the availability of tonometry measures in 2 cohorts nested within the same study using the same measurement protocol. This allowed a large-scale exploration of various tonometry-based measures of arterial stiffness. Our study provides further insight into the relative contributions of various markers of glucose dysregulation to arterial stiffness.

There are limitations of our investigation. First, the cross-sectional design precludes any causal inference about the association of dysglycemia and arterial stiffness, especially as the relation of dysglycemia and arterial stiffness can be bidirectional.^{17,33} Hence, it remains to be investigated whether insulin resistance or dysglycemia will prospectively increase arterial stiffness. Additionally, individuals with diabetes should be examined prospectively to determine the impact of the joint association of arterial stiffness and diabetes on the risk of incident hypertension and CVD, a premise that we did not evaluate. Second, our study sample consisted of relatively healthy young-to-middle-aged community-dwelling White individuals, limiting the generalizability of our observations to other age groups and ethnicities.

In conclusion, among adults, abnormal markers of glucose homeostasis were consistently associated with abnormal arterial stiffness indices, including CFPWV and Alx. The observed associations potentially reflect the high burden of a subclinical precursor of future CVD

among individuals with dysglycemia, a premise that warrants examination in future investigations. Furthermore, the usefulness of arterial tonometry measures for CVD risk stratification in the setting of dysglycemia merits further investigation.

PERSPECTIVES

A growing body of evidence links dysglycemia and insulin resistance to the development of CVD, yet the vascular pathways through which these metabolic abnormalities exert risk are incompletely understood. Arterial stiffness is a well-established marker of subclinical vascular damage and an independent predictor of cardiovascular events, making it a clinically relevant intermediate phenotype. Using data from a large, community-based cohort, our findings demonstrate that adverse levels of glycemic markers and insulin resistance measures are consistently associated with abnormalities in central arterial stiffness, even in a relatively young population with a low prevalence of diabetes.

These observations support the concept that vascular changes related to impaired glucose metabolism begin early, before overt diabetes, and may contribute to excess cardiovascular risk across the glycemic spectrum. The differential associations observed across stiffness measures highlight the complexity of arterial remodeling and suggest that glycemia and insulin resistance may have heterogeneous effects on central versus peripheral vasculature. Importantly, the associations persisted after adjustment for multiple cardiovascular risk factors, underscoring the independent vascular relevance of metabolic dysfunction.

From a clinical and public health perspective, these data reinforce the importance of early identification and management of dysglycemia to preserve vascular health. Future longitudinal studies are needed to determine whether improvement in glycemic control and insulin sensitivity translates into favorable changes in arterial stiffness and, ultimately, reductions in cardiovascular events.

ARTICLE INFORMATION

Received November 6, 2025; accepted June 22, 2026.

Affiliations

Department of Medicine, University of Maryland Midtown Campus, Baltimore (S.R.N.). Division of General Medicine, Section for Research, Beth Israel Deaconess Medical Center, Boston, MA (S.P.J.). Renal-Electrolyte and Hypertension Division (J.B.C.) and Department of Biostatistics, Epidemiology, and Informatics (J.B.C.), Perelman School of Medicine, University of Pennsylvania, Philadelphia. Department of Medicine, Division of Endocrinology, Diabetes & Metabolism, Johns Hopkins School of Medicine, Baltimore, MD (J.B.E.-T.). Welch Prevention Center for Prevention, Epidemiology and Clinical Research, Johns Hopkins University, Baltimore, MD (J.B.E.-T.).

Acknowledgments

The authors thank the staff and participants of the Framingham Heart Study.

Author Contributions

J.B. Echouffo-Tcheugui had full access to all the data in the study and takes responsibility for the integrity of the data and the accuracy of the data analyses. The study concept and design were contributed by J.B. Echouffo-Tcheugui. All authors contributed to the acquisition, analysis, or interpretation of data. S.R. Noumegni contributed to drafting the article. All authors contributed to the critical revision of the article for important intellectual content. S.R. Noumegni and J.B. Echouffo-Tcheugui contributed to the statistical analysis. J.B. Echouffo-Tcheugui contributed to the administrative, technical, or material support. J.B. Echouffo-Tcheugui contributed to study supervision.

Sources of Funding

J.B. Echouffo-Tcheugui was supported by the National Institutes of Health/National Heart, Lung, and Blood Institute grant K23 HL153774. S.R. Noumegni was supported by the French Federation of Cardiology, the Philippe Foundation, and the Servier Institute.

Disclosures

None.

Supplemental Material

Tables S1–S11

Figure S1

REFERENCES

1. Sarwar N, Gao P, Seshasai SRK, Gobin R, Kaptoge S, Di Angelantonio E, Ingelsson E, Lawlor DA, Selvin E, Stampfer M, et al; Emerging Risk Factors Collaboration. Diabetes mellitus, fasting blood glucose concentration, and risk of vascular disease: a collaborative meta-analysis of 102 prospective studies. *Lancet*. 2010;375:2215–2222. doi: 10.1016/S0140-6736(10)60484-9
2. Sowers JR. Diabetes mellitus and vascular disease. *Hypertension*. 2013;61:943–947. doi: 10.1161/HYPERTENSIONAHA.111.00612
3. Low Wang CC, Hess CN, Hiatt WR, Goldfine AB. Clinical update: cardiovascular disease in diabetes mellitus: atherosclerotic cardiovascular disease and heart failure in type 2 diabetes mellitus - mechanisms, management, and clinical considerations. *Circulation*. 2016;133:2459–2502. doi: 10.1161/CIRCULATIONAHA.116.022194
4. Wilkinson IB, Maki-Petäjä KM, Mitchell GF. Uses of arterial stiffness in clinical practice. *Arterioscler Thromb Vasc Biol*. 2020;40:1063–1067. doi: 10.1161/ATVBAHA.120.313130
5. Prentner SB, Chirinos JA. Arterial stiffness in diabetes mellitus. *Atherosclerosis*. 2015;238:370–379. doi: 10.1016/j.atherosclerosis.2014.12.023
6. Mitchell GF. Arterial stiffness and hypertension: chicken or egg? *Hypertension*. 2014;64:210–214. doi: 10.1161/HYPERTENSIONAHA.114.03449
7. Kannel WB, Feinleib M, McNamara PM, Garrison RJ, Castelli WP. An investigation of coronary heart disease in families. The Framingham offspring study. *Am J Epidemiol*. 1979;110:281–290. doi: 10.1093/oxfordjournals.aje.a112813
8. Splansky GL, Corey D, Yang Q, Atwood LD, Cupples LA, Benjamin EJ, D'Agostino RB, Fox CS, Larson MG, Murabito JM, et al. The Third Generation Cohort of the National Heart, Lung, and Blood Institute's Framingham Heart Study: design, recruitment, and initial examination. *Am J Epidemiol*. 2007;165:1328–1335. doi: 10.1093/aje/kwm021
9. American Diabetes Association Professional Practice Committee for Diabetes. 2. Diagnosis and classification of diabetes: standards of care in diabetes-2026. *Diabetes Care*. 2026;49:S27–S49. doi: 10.2337/dc26-S002
10. Matthews DR, Hosker JP, Rudenski S, Naylor B, Treacher DF, Turner RC. Homeostasis model assessment: insulin resistance and beta-cell function from fasting plasma glucose and insulin concentrations in man. *Diabetologia*. 1985;28:412–419. doi: 10.1007/BF00280883
11. Katz A, Nambi SS, Mather K, Baron AD, Follmann DA, Sullivan G, Quon MJ. Quantitative insulin sensitivity check index: a simple, accurate method for assessing insulin sensitivity in humans. *J Clin Endocrinol Metab*. 2000;85:2402–2410. doi: 10.1210/jcem.85.7.6661
12. Muniyappa R, Lee S, Chen H, Quon MJ. Current approaches for assessing insulin sensitivity and resistance in vivo: advantages, limitations, and appropriate usage. *Am J Physiol Endocrinol Metab*. 2008;294:E15–E26. doi: 10.1152/ajpendo.00645.2007
13. Wallace TM, Levy JC, Matthews DR. Use and abuse of HOMA modeling. *Diabetes Care*. 2004;27:1487–1495. doi: 10.2337/diacare.27.6.1487
14. Mitchell GF, Hwang S-J, Vasan RS, Larson MG, Pencina MJ, Hamburg NM, Vita JA, Levy D, Benjamin EJ. Arterial stiffness and cardiovascular

events: the Framingham Heart Study. *Circulation*. 2010;121:505–511. doi: 10.1161/CIRCULATIONAHA.109.886655

15. Vasan RS, Pan S, Xanthakis V, Beiser A, Larson MG, Seshadri S, Mitchell GF. Arterial stiffness and long-term risk of health outcomes: the Framingham Heart Study. *Hypertension*. 2022;79:1045–1056. doi: 10.1161/HYPERTENSIONAHA.121.18776
16. Kumrular M, Karaca Ozer P, Elitok A. The role of aortic stiffness parameters in evaluating myocardial ischemia. *Cardiol Res*. 2020;11:328–336. doi: 10.14740/cr1125
17. Cohen JB, Mitchell GF, Gill D, Burgess S, Rahman M, Hanff TC, Ramachandran VS, Mutalik KM, Townsend RR, Chirinos JA. Arterial stiffness and diabetes risk in Framingham Heart Study and UK Biobank. *Circ Res*. 2022;131:545–554. doi: 10.1161/CIRCRESAHA.122.320796
18. Ji H, Kwan AC, Chen MT, Ouyang D, Ebinger JE, Bell SP, Niiranen TJ, Bello NA, Cheng S. Sex differences in myocardial and vascular aging. *Circ Res*. 2022;130:566–577. doi: 10.1161/CIRCRESAHA.121.319902
19. Salomaa V, Riley W, Kark JD, Nardo C, Folsom AR. Non-insulin-dependent diabetes mellitus and fasting glucose and insulin concentrations are associated with arterial stiffness indexes. The ARIC Study. *Atherosclerosis Risk in Communities Study*. *Circulation*. 1995;91:1432–1443. doi: 10.1161/01.cir.91.5.1432
20. Webb DR, Khunti K, Silverman R, Gray LJ, Srinivasan B, Lacy PS, Williams B, Davies MJ. Impact of metabolic indices on central artery stiffness: independent association of insulin resistance and glucose with aortic pulse wave velocity. *Diabetologia*. 2010;53:1190–1198. doi: 10.1007/s00125-010-1689-9
21. Liang J, Zhou N, Teng F, Zou C, Xue Y, Yang M, Song H, Qi L. Hemoglobin A1c levels and aortic arterial stiffness: the Cardio-metabolic Risk in Chinese (CRC) study. *PLoS One*. 2012;7:e38485. doi: 10.1371/journal.pone.0038485
22. Lee YH, Shin MH, Choi JS, Rhee JA, Nam HS, Jeong SK, Park KS, Ryu SY, Choi SW, Kim BH, et al. HbA1c is significantly associated with arterial stiffness but not with carotid atherosclerosis in a community-based population without type 2 diabetes: the Dong-gu study. *Atherosclerosis*. 2016;247:1–6. doi: 10.1016/j.atherosclerosis.2016.01.032
23. Shin JY, Lee HR, Lee DC. Increased arterial stiffness in healthy subjects with high-normal glucose levels and in subjects with pre-diabetes. *Cardio-vasc Diabetol*. 2011;10:30. doi: 10.1186/1475-2840-10-30
24. Gomez-Sanchez L, Garcia-Ortiz L, Patino-Alonso MC, Recio-Rodriguez JI, Feuerbach N, Marti R, Agudo-Conde C, Rodriguez-Sanchez E, Maderuelo-Fernandez JA, Ramos R, et al; MARK Group. Glycemic markers and relation with arterial stiffness in Caucasian subjects of the MARK study. *PLoS One*. 2017;12:e0175982. doi: 10.1371/journal.pone.0175982
25. Markus MRP, Rosaleszcz S, Iltermann T, Baumeister SE, Schipf S, Siewert-Markus U, Lorbeer R, Storz C, Ptushkina V, Peters A, et al. Glucose and insulin levels are associated with arterial stiffness and concentric remodeling of the heart. *Cardiovasc Diabetol*. 2019;18:145. doi: 10.1186/s12933-019-0948-4
26. Eguchi K, Hoshida S, Miyashita H, Nagasaka S, Kario K. Why the radial augmentation index is low in patients with diabetes: the J-HOP study. *Atherosclerosis*. 2016;246:338–343. doi: 10.1016/j.atherosclerosis.2016.01.034
27. Climie RED, Nikolic SB, Otahal P, Keith LJ, Sharman JE. Augmentation index and arterial stiffness in patients with type 2 diabetes mellitus. *Artery Res*. 2013;7:194–200. doi: 10.1016/j.artres.2013.09.002
28. Cheng L-T, Tang L-J, Cheng L, Huang H-Y, Wang T. Limitation of the augmentation index for evaluating arterial stiffness. *Hypertens Res*. 2007;30:713–722. doi: 10.1291/hypres.30.713
29. Rabkin SW, Chan SH, Sweeney C. Ankle-brachial index as an indicator of arterial stiffness in patients without peripheral artery disease. *Angiology*. 2012;63:150–154. doi: 10.1177/0003319711410307
30. Sengstock DM, Vaitkevicius PV, Supiano MA. Arterial stiffness is related to insulin resistance in nondiabetic hypertensive older adults. *J Clin Endocrinol Metab*. 2005;90:2823–2827. doi: 10.1210/jc.2004-1686
31. Ikonomidis I, Lambadiari V, Pavlidis G, Koukoulis C, Kousathana F, Varoudi M, Spanoudi F, Maratou E, Parissis J, Triantafyllidi H, et al. Insulin resistance and acute glucose changes determine arterial elastic properties and coronary flow reserve in dysglycaemic and first-degree relatives of diabetic patients. *Atherosclerosis*. 2015;241:455–462. doi: 10.1016/j.atherosclerosis.2015.06.006
32. Murai N, Saito N, Nii S, Nishikawa Y, Suzuki A, Kodama E, Iida T, Mikura K, Imai H, Hashizume M, et al. Postloading insulinemia is independently associated with arterial stiffness in young Japanese persons. *Hypertens Res*. 2021;44:1515–1523. doi: 10.1038/s41440-021-00749-4
33. Zheng M, Zhang X, Chen S, Song Y, Zhao D, Gao X, Wu S. Arterial stiffness preceding diabetes: a longitudinal study. *Circ Res*. 2020;127:1491–1498. doi: 10.1161/CIRCRESAHA.120.317950
34. Ikonomidis I, Pavlidis G, Thymis J, Birba D, Kalogeris A, Kousathana F, Kountouri A, Balamanis K, Parissis J, Andreadou I, et al. Effects of glucagon-like peptide-1 receptor agonists, sodium-glucose cotransporter-2 inhibitors, and their combination on endothelial glycocalyx, arterial function, and myocardial work index in patients with type 2 diabetes mellitus after 12-month treatment. *J Am Heart Assoc*. 2020;9:e015716. doi: 10.1161/JAHA.119.015716

FIRST PROOF ONLY